Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

Course Introduction
Video • 1 min

Course Overview
Reading • 2 min

Syllabus
Reading • 10 min

Meet Your Instructor
Video • 1 min

Meet Your Fellow Learners
Discussion Prompt • 10 min

Academic Integrity
Reading • 1 min

Lesson 1: Introduction to Linear Algebra

Introduction to Machine Learning
Reading • 25 min

Introduction to Linear Algebra
Reading • 4 min

Why Linear Algebra and Mathematics?
Reading • 2 min

Notation
Reading • 5 min

Lesson 2: Systems of Linear Equations

Foundational Concepts of Systems of Linear Equations
Reading • 20 min

[H5P] System of Linear Equations
Ungraded App Item • 20 min

Solved Example and Recommended Resources
Reading • 20 min

Check Your Knowledge: System of Linear Equations
Graded Assignment • 16 min

Lesson 3: Matrices

Matrices and Matrix Operations
Reading • 20 min

Matrices
Video • 8 min

Helpful Resources and Solved Examples
Reading • 10 min

Check Your Knowledge: Matrices
Graded Assignment • 16 min

Lesson 4: Vector Spaces

Foundations of Vector Spaces: Operations and Subspaces
Reading • 20 min

Vector Space
Video • 7 min

Vector Space Properties
Reading • 10 min

Subspace
Reading • 1 min

Check Your Knowledge: Vector Spaces
Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

Introduction to Orthogonal Complement
Reading • 20 min

Introduction to Orthogonal Projections
Reading • 40 min

Foundations of Vector Spaces: Operations and Subspaces

0:05 / 2:01 2x

Read Section 2.4: "Vector Space" from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this section, you will explore the concept of vector spaces. A vector space is a set equipped with two operations: vector addition (inner operation) and scalar multiplication (outer operation). These operations must satisfy specific properties such as closure, associativity, and distributivity.

The section begins by discussing groups, an essential algebraic structure that supports an understanding of the operations on sets. From there, it extends these ideas to define vector spaces, introducing the necessary conditions that make a set with vector addition and scalar multiplication a vector space. The section also discusses vector subspaces, which are subsets of a vector space that themselves satisfy the conditions of a vector space.

As you read, consider the following questions:

1. How do the properties of a group help in defining a vector space, and why are these properties essential for the structure of a vector space?
2. In what ways do vector spaces extend the concept of groups, and how does the introduction of scalar multiplication impact the structure and operations within a vector space?
3. Consider a vector space V and a subset U of V. What are the necessary conditions for U to be a subspace of V, and why are these conditions crucial for maintaining the structure of a vector space within the subspace?
4. What are the implications of the closure property in the context of vector subspaces? How does this property ensure that operations performed within a subspace do not lead to elements outside of the subspace?
5. In practical applications, how do vector subspaces aid in dimensionality reduction, and what are the benefits of working within a subspace rather than the entire vector space?

Engage with these questions to deepen your understanding of the key concepts and structures introduced in this section.

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